

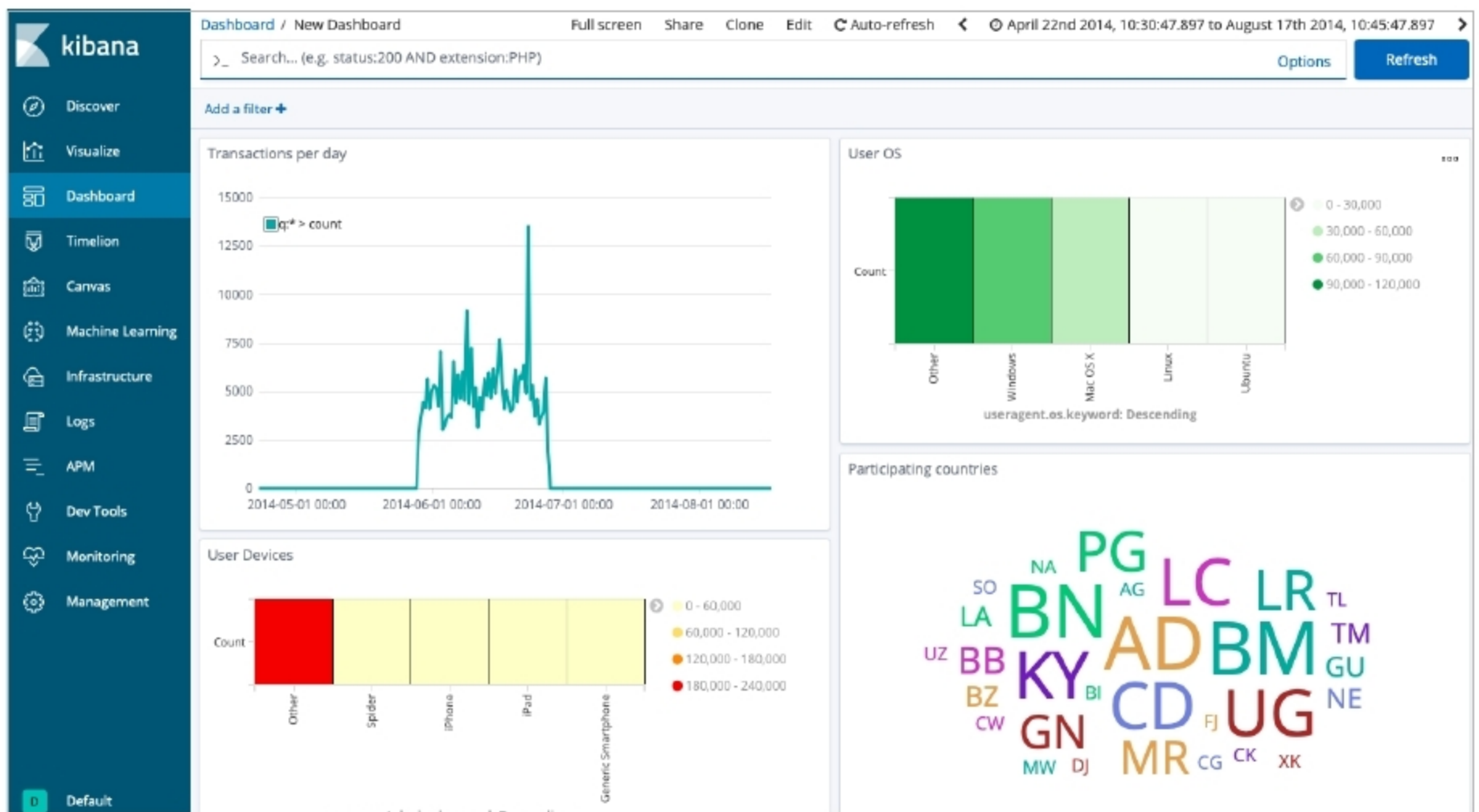


Figure 8: A dashboard for visualisations in Kibana

the help of `_id`. The document is saved to the primary shard. If successful, the request is forwarded to the replica nodes in parallel and while inserting a document, Elasticsearch will locate the primary shard for the index and insert the document there.

- **Search:** The related shard will be retrieved based on the following formula. Using round robin scheduling on the primary or replica nodes, one is chosen to carry out the current query. Round robin scheduling between primary and replicated nodes optimises the load on the search engine. The explained architecture uses round robin scheduling to balance the load on nodes.
- **Delete:** The *delete* request follows similar steps as inserting. The master node finds the associated primary shard and deletes the document from there. Afterwards, it will also be deleted from replicated shards.

3. Kibana

Kibana is a tool to visualise Elasticsearch data in real-time. It provides a variety of visualisations to get the best meaning out of your data. Dashboards can be created according to use cases, and get updated with new data automatically.

Kibana is a visualisation tool built on top of Elasticsearch, which connects to the latter using Elasticsearch's REST APIs. It helps us understand data better from a non-developer perspective by providing various forms such as tables, rows, charts, geo maps, heat maps, and so on.

Some of the main features of Kibana are:

- Leverages the aggregation capabilities of Elasticsearch
- Has many predefined visualisations and it is very easy to define new

ones using Vega

- Can plot complex data like geo-data, time series metrics, relationship graphs, etc
- Unsupervised machine learning has been added to detect anomalies in Elasticsearch data
- Dashboards can be made and transported in the form of PDF, PNG or level-controlled links
- Alerts and notifications can be set up for any specific query

The ELK stack helps users to collect data from various sources, enhance it and store it in a self-replicating distributed manner. Stored data can be accessed in real-time and can be presented to the user in various forms such as pie-charts or geo-mapping, with the help of Kibana. It helps enterprises in collecting data to provide it to businesses and users in real-time. **END** 🐧

By: Vaibhav Aggarwal and Prof. B. Thangaraju

The authors are associated with the Open Source Technology Lab at the International Institute of Information Technology, Bengaluru.